DECK 16 · WEEK 8 · MORNING 1

Fixed Income IV
Credit & Securitization

How credit risk is measured, analysed across issuer types, and repackaged into asset-backed securities

CFA Level I · Vol 6 Learning Modules 14–19

Roadmap — Fixed Income IV — Credit Analysis & Securitization

5 sections · ~2 hours · Built bottom-up from CFA readings

01

Credit-risk framework

POD · LGD · EAD · EL = POD × LGD · recovery rates · rating scales · spread-driven HPR

02

Sovereign credit analysis

Ability vs willingness to pay · fiscal strength · growth · external stability · Moody's ratio bands

03

Corporate credit analysis

4 C's · EBIT margin / coverage / leverage · seniority · priority of claims · structural subordination

04

Securitization structures

SPV · true-sale bankruptcy-remote · tranching · credit enhancement · non-mortgage ABS · CLOs · covered bonds

05

MBS specifics

RMBS vs CMBS · agency vs non-agency · WAC/WAM · prepayment · SMM/CPR/PSA · sequential-pay CMO · PAC · balloon risk

Learning objectives

By the end of this deck you should be able to:

1. Decompose credit risk into probability of default, loss given default, and exposure at default, and compute expected loss in currency or percentage terms.

2. Interpret the long-term rating scales of Moody's, S&P and Fitch and explain the investment-grade vs high-yield distinction and the role of rating migration.

3. Evaluate a sovereign issuer on both qualitative (institutions, willingness) and quantitative (debt/GDP, external reserves, current account) dimensions.

4. Apply the 4 C's to a corporate issuer, compute key ratios (EBIT margin, EBIT/interest, Debt/EBITDA, RCF/Net Debt), and rank claims in a bankruptcy waterfall.

5. Describe securitization mechanics — SPV, true sale, tranching, internal and external credit enhancement — and work out loss absorption across tranches.

6. Analyse MBS cash flows using WAC, WAM, SMM, CPR and the PSA benchmark, and distinguish sequential-pay CMOs, PAC tranches, and CMBS balloon risk.

SECTION 01

Credit-risk framework (PD × LGD × EAD)

Credit risk is the expected economic loss from a borrower's failure to pay — decomposable into probability of default, loss given default, and exposure at default

POD is a forward-looking annual probability; LGD combines severity (1 − RR) with claim size (EAD). Their product is the expected loss per period.

$$EL = POD \times LGD \qquad \text{where} \quad LGD = EE \times (1 - RR)$$

where: POD = probability of default (annualized); EE = expected exposure (EAD); RR = recovery rate; LGD = loss given default

1. **Probability of default (POD)**
   Forward-looking likelihood — typically annualized — that the issuer fails to make full & timely payments. Driven by profitability, coverage and leverage.

2. **Loss given default (LGD)**
   Severity of loss if default occurs = EE × (1 − RR). Secured creditors recover more; deeply subordinated debt often has LGD ≈ 100%.

3. **Expected exposure (EAD / EE)**
   Claim size at time of default = face + accrued interest − market value of available collateral. Drives the currency amount of loss.

Source: CFA Institute, Curriculum Vol 6, Learning Module 14, pp. 341–350.

Expected loss equals probability of default times loss given default — a fair credit spread approximates this same product

Collateral reduces EAD before the RR haircut — so a 90% recovery on an already-secured loan produces a very small expected loss.

$$EL = POD \times (EE - Collateral) \times (1 - RR) \qquad \text{Credit Spread} \approx POD \times LGD$$

where: Fair-spread approximation: investor is compensated for credit risk if the spread ≥ annualized expected loss.

CFA LM 14 SELF-ASSESSMENT Q2 — EUR 500,000 SECURED LOAN

POD = 5%, expected exposure EUR 400,000, collateral EUR 100,000, recovery rate on unrecovered claim = 90%.

1. Net exposure after collateral = 400,000 − 100,000 = EUR 300,000
2. Severity = 1 − RR = 1 − 0.90 = 10%
3. LGD in currency = 300,000 × 10% = EUR 30,000
4. EL = POD × LGD = 0.05 × 30,000 = EUR 1,500

EL = EUR 1,500 (0.30% of face). If spread > 30 bps, investors are more than fairly compensated.

Source: CFA Institute, Curriculum Vol 6, Learning Module 14, Self-Assessment Q2, pp. 342–354.

The three major rating agencies split the debt universe at Baa3 / BBB– — below that line, bonds are high-yield (speculative-grade)

Credit-migration (downgrade) risk matters even without default — an IG-to-HY "fallen-angel" downgrade can force forced selling by mandate-constrained funds.

Grade band	Moody's	S&P	Fitch	Description
Highest quality	Aaa	AAA	AAA	Lowest credit risk
Very high quality	Aa1 – Aa3	AA+ – AA–	AA+ – AA–	Very low credit risk
High quality	A1 – A3	A+ – A–	A+ – A–	Low credit risk
Lower medium — IG floor	Baa1 – Baa3	BBB+ – BBB–	BBB+ – BBB–	Moderate credit risk
Speculative	Ba1 – Ba3	BB+ – BB–	BB+ – BB–	Substantial credit risk
Highly speculative	B1 – B3	B+ – B–	B+ – B–	High credit risk
Substantial risk / near default	Caa1 – Ca	CCC+ – CC	CCC – CC	Default a real possibility

THE BBB– / Baa3 LINE

Above = investment grade (institutional mandates, low spreads, low default rates historically <1% p.a.). Below = high yield: covenants tighter, secured financing more common, spreads 300–800 bps+ typical.

Source: CFA Institute, Curriculum Vol 6, Learning Module 14, Exhibit 7, pp. 355–360.

Key financial ratios drive both POD (profitability, coverage, leverage) and LGD (seniority) — and hence credit ratings

Rating ≠ a single ratio — agencies combine multiple metrics, industry benchmarks and qualitative factors. Bands below are generic investment-grade corporate indicative.

Moody's rating	EBIT margin	EBIT / Interest	Debt / EBITDA	CF / Net Debt
Aaa	> 25%	> 25×	< 1.0×	> 80%
Aa	20 – 25%	15 – 25×	1.0 – 1.5×	60 – 80%
A	15 – 20%	10 – 15×	1.5 – 2.5×	40 – 60%
Baa (IG floor)	10 – 15%	5 – 10×	2.5 – 3.5×	25 – 40%
Ba	7 – 10%	3 – 5×	3.5 – 5.0×	15 – 25%
B	3 – 7%	1.5 – 3×	5.0 – 7.0×	5 – 15%
Caa and below	< 3%	< 1.5×	> 7.0×	< 5%

READING THE TABLE

POD drivers: higher EBIT margin + higher coverage + lower leverage = lower POD = higher rating. LGD is set separately by seniority (next slides).

Source: CFA Institute, Curriculum Vol 6, Learning Module 14, Exhibit 5, pp. 352–354.

Credit spreads widen at the bottom of the credit cycle and narrow at the top — spread change and duration together drive holding-period return

Same mechanics as interest-rate risk: spread ΔY acts on price through modified duration, with a small convexity correction.

$$\%\Delta PV_full = -(AnnModDur \times \Delta Spread) + \tfrac{1}{2} \times AnnConvexity \times (\Delta Spread)^2$$

where: Spread narrowing ⇒positive price return; spread widening ⇒negative. Convexity is typically small but positive for plain bonds.

CFA LM 14 SELF-ASSESSMENT Q6 — HY BOND, +100 BPS SPREAD WIDENING IN A RECESSION

Annualized modified duration = 4.5; reported annual convexity = 0.23 (rescale to 23 to pair with $\%^2$); spread shock ΔS = +1.00%.

1. Duration effect: $-4.5 \times 1.00\% = -4.500\%$
2. Convexity effect: $\tfrac{1}{2} \times 23 \times (1.00\%)^2 = \tfrac{1}{2} \times 23 \times 0.0001 \times 10{,}000 = +0.115\%$
3. (Using % units throughout: $23 \times 0.5 \times 1.0^2 = 11.5$ bps = 0.115%)
4. $\%\Delta PV_full = -4.500\% + 0.115\% = -4.385\%$

%ΔPV_full ≈ −4.385% — convexity softens the blow by ~12 bps. Spread narrowing of the same size would produce +4.615%.

Source: CFA Institute, Curriculum Vol 6, Learning Module 14, Self-Assessment Q6, pp. 362–372.

SECTION 02

Sovereign credit analysis

Sovereign creditworthiness hinges on BOTH ability to pay (taxation, FX reserves) AND willingness to pay — sovereign immunity blocks forced bankruptcy

Qualitative framework used by all three major rating agencies, with variants. Willingness is harder to observe than ability — examples: Uruguay 2003, Argentina 2001.

GOVERNMENT INSTITUTIONS

Rule of law · political stability · willingness-to-pay track record · transparency of data.

FISCAL FLEXIBILITY

Tax base · ability to adjust revenue & expenditure · track record of fiscal discipline.

MONETARY EFFECTIVENESS

Central bank independence · credibility · exchange-rate regime · bond-market depth.

ECONOMIC FLEXIBILITY

Size · per-capita income · diversification · growth · shock absorption capacity.

EXTERNAL STATUS

Reserve-currency status · access to external funding · geopolitical risk · FX flexibility.

WILLINGNESS MATTERS — URUGUAY 2003

After the banking crisis & peso devaluation, Uruguay carried out an IMF-supported voluntary debt exchange that extended maturities by 5 years. Although "voluntary," analysts deemed it a default-in-substance — a reminder that sovereigns can subordinate holdouts under sovereign-immunity protection.

Source: CFA Institute, Curriculum Vol 6, Learning Module 15, Exhibit 1–2, pp. 380–388.

Quantitative sovereign analysis centres on three ratio families: fiscal strength, economic growth and stability, and external stability

Numerator is usually debt or interest; denominator uses GDP (ability to tax), Revenue (fiscal resources), or FX reserves (external firepower).

Family	Ratio	Definition	Aaa	Ba	Caa
Fiscal — debt burden	Debt / GDP	General govt debt to GDP	< 35%	60–90%	> 120%
Fiscal — debt burden	Debt / Revenue	General govt debt to govt revenue	< 100%	250–400%	> 500%
Fiscal — affordability	Interest / GDP	Interest payments to GDP	< 1%	4–8%	> 15%
Fiscal — affordability	Interest / Revenue	Interest payments to govt revenue	< 5%	15–25%	> 40%
Growth & stability	Real GDP growth	Avg growth, past decade	2–4%	1–3%	negative
External stability	FX reserves / GDP	Foreign currency reserves	> 15%	5–10%	< 3%
External stability	Reserve ratio	FX reserves / external debt	> 80%	30–50%	< 10%

Source: CFA Institute, Curriculum Vol 6, Learning Module 15, Exhibits 4–7, pp. 387–394.

Illustrative peer comparison shows how Moody's differentiates an A3 Malaysia from a Ba3 Vietnam and a B3 Moldova on the same four metrics

Reserve-currency issuers (USD, EUR, JPY, GBP, CHF) sit in a different universe — structural deficits and high debt/GDP do not mechanically downgrade them.

Sovereign	Moody's	Debt / GDP	Real GDP growth	FX Reserves / GDP	Reserve ratio
Malaysia	A3	~65%	4.5%	11.9%	54%
Thailand	Baa1	~58%	3.0%	16.4%	102%
Philippines	Baa2	~58%	6.0%	10.3%	96%
Vietnam	Ba3	~42%	6.4%	9.9%	84%
Indonesia	Baa2	~40%	5.2%	3.9%	31%
Moldova	B3	~35%	2.0%	20% (IMF)	IMF dep.

RESERVE-CURRENCY vs NON-RESERVE SOVEREIGNS

A reserve-currency sovereign (US, Eurozone, UK, Japan, Switzerland) can borrow in its own currency from foreign investors and monetize if necessary — so debt/GDP ratios of 100%+ can still sit at AA–AAA. Non-reserve sovereigns are capped by FX reserve adequacy and external-debt discipline.

[1] Ratios are indicative FYE 2020–2021 values used in the CFA learning examples — illustrative, not current.

[2] Malaysia A3, Thailand Baa1, Philippines Baa2, Vietnam Ba3, Moldova B3 (Moody's, indicative at time of writing).

Source: CFA Institute, Curriculum Vol 6, Learning Module 15, Examples 4, 6, 7, pp. 385–393.

SECTION 03

Corporate credit analysis

Traditional credit analysis uses the "Cs of credit" — five bottom-up factors (Capacity, Capital, Collateral, Covenants, Character) and three top-down (Country, Conditions, Currency)

Bottom-up Cs drive the issuer's own POD/LGD. Top-down Cs describe the macro environment faced by ALL borrowers in a jurisdiction.

COUNTRY

Geopolitical + legal environment. Rule of law, political stability, bankruptcy-code efficiency, property rights.

CONDITIONS

General economic, competitive, business environment. Where we are in the credit cycle.

CURRENCY

FX exposure of CFs. Cross-border borrowers with FX-debt or imports face currency mismatch risk.

CAPACITY

Ability to service debt on time (profitability, CFs).

CAPITAL

Other resources reducing debt reliance (equity, liquidity).

COLLATERAL

Quality & value of pledged assets supporting debt.

COVENANTS

Legal terms restricting issuer behaviour.

CHARACTER

Management quality & willingness to pay.

QUANTITATIVE vs QUALITATIVE

Capacity & Capital are mostly quantitative (ratios). Collateral, Covenants, Character are largely qualitative — requiring analyst judgment on management track record and covenant strength.

Source: CFA Institute, Curriculum Vol 6, Learning Module 14, Exhibit 1, pp. 344–346.

Corporate POD is driven by profitability, coverage and leverage — higher profits and coverage with lower leverage mean lower POD

EBIT vs EBITDA vs FFO vs RCF — be explicit about which cash-flow measure is in numerator/denominator before comparing across issuers.

Family	Ratio	Formula	Higher =
Profitability	EBIT margin	EBIT / Revenue	Better
Profitability	EBITDA margin	EBITDA / Revenue	Better
Coverage	EBIT / Interest	EBIT / Interest expense	Better
Coverage	EBITDA / Interest	(EBIT + D&A) / Interest expense	Better
Coverage — full burden	EBITDAR / (Int + Rent)	(EBITDA + Rent) / (Interest + Rent)	Better
Leverage	Debt / EBITDA	Total debt / EBITDA	Worse
Leverage	Debt / Capital	Debt / (Debt + Equity)	Worse
Leverage — cash-flow based	FFO / Debt	Funds from operations / Debt	Better
Leverage — cash-flow based	RCF / Net Debt	(FFO – Dividends) / (Debt – Cash)	Better

Source: CFA Institute, Curriculum Vol 6, Learning Module 16, Exhibit 3, pp. 417–420.

A downside-scenario ratio walk shows how a compressing EBIT margin propagates to weaker coverage and higher leverage — credit-spread widening follows

Bowstream: mid-sized IG consumer-durables issuer. Stress case — input cost inflation and rising rates, margin down 20 ppt, debt growing 10% p.a.

Metric	Y0 (A)	Y1	Y2	Y3	Y4	Y5	Rating drift
EBIT margin	25%	20%	15%	10%	5%	5%	A → Ba
EBIT / Interest	51.9×	32.9×	23.0×	14.4×	5.8×	5.2×	Aaa→ Baa
Debt / EBITDA	0.36×	0.43×	0.54×	0.71×	1.06×	1.21×	Aaa→ Aa
RCF / Net Debt	379%	271%	191%	130%	67%	54%	Aaa→ Baa
Implied credit outlook	Stable	Stable	Neg.	Neg.	Downgrade	Downgrade	—

TAKEAWAY — RATINGS ARE PROCYCLICAL

By year 4, coverage, leverage and CF metrics have all crossed Baa-boundary thresholds. In practice: agency would put Bowstream on negative watch between Y2–Y3 and downgrade by Y4. Spread widening typically precedes the ratings action by 1–2 quarters.

Source: CFA Institute, Curriculum Vol 6, Learning Module 16, Example 4 (Bowstream), pp. 419–423.

Seniority ranking determines how much an investor recovers in default — highest for first-lien secured, near-zero for deeply subordinated

Typical NA non-financial recovery rates (Moody's, 2000–2019 avg). Actual recoveries vary by cycle — 2008–09 trough recoveries were ~50% of long-run averages.

Rank	Class	Claim type	Typical recovery	LGD
1	First lien / first mortgage	Pledged specific assets (plant, inventory)	~65%	35%
2	Senior secured	Pledged broader collateral pool	~55%	45%
3	Junior secured / second lien	Residual claim on pledged assets after 1st lien	~45%	55%
4	Senior unsecured	General claim on unpledged assets	~35%	65%
5	Senior subordinated	Subordinated but senior to sub debt	~25%	75%
6	Subordinated	Below all senior unsecured	~15%	85%
7	Junior subordinated	Deeply subordinated, often LGD ≈ 100%	~5%	95%
8	Equity	Residual; wiped out before unsecured creditors	0%	100%

NOTCHING

Issuer rating = rating of senior unsecured debt. Secured debt typically notched UP 1 notch; subordinated debt notched DOWN 1–2 notches.

Source: CFA Institute, Curriculum Vol 6, Learning Module 16, Exhibits 4–5, pp. 425–432.

On default the priority of claims cascades through a strict waterfall — a small pool of pledged collateral can still leave subordinated holders wiped out

Mojofon (illustrative): insolvency, recovery pool of 150. Secured claims ride the collateral; senior unsec share the residual; sub is behind them.

Claim	Face	Recovery	RR
Senior secured (Coll 60)	50	50	100%
Senior unsecured	35	33.3	~95%
Subordinated	15	0	0%
Common equity	100	0	0%
Pool available	**200**	**150**	**—**

WATERFALL — RECOVERY IN DEFAULT

Insolvency estate is 150. Secured bonds (50) are collateralised by assets worth 60 (excess 10 flows to unsecured pool).

1. Step 1 — Secured: 50 face × 100% recovery (collateral worth 60 ≥ 50) = 50
2. Step 2 — Excess collateral of 10 is released into the unsecured pool
3. Step 3 — Pool for unsecured = 150 − 50 + 10 = 110, of which non-sec claims = 50 + 35 = unsec total
4. Actually: residual pool after secured = 100. Sr-unsec 35 / Sub 15 = ranks differently
5. Sr-unsec get their 35 in full first; Sub 15 = 15; leftover to equity = 50

Realistic split depends on estate size. Key principle: pari-passu within a class; strict priority across classes.

Source: CFA Institute, Curriculum Vol 6, Learning Module 16, Practice Problems 4–8 (Mojofon), pp. 433–437.

Structural subordination arises when holding-company debt is serviced only after operating-subsidiary debt — subsidiary bonds recover more than HoldCo bonds even if pari passu on paper

Private-equity LBO structures frequently exploit this: push operating CFs into a subsidiary, issue lightly-secured HoldCo PIK notes above to amplify equity returns.

HOLDING-CO DEBT

HoldCo owns stock of OpCo. HoldCo debt is paid from dividends UP-streamed by OpCo — but only AFTER OpCo has paid its own creditors.

In default: HoldCo creditors hold equity claim on OpCo, which is BELOW OpCo creditors.

Result: HoldCo senior unsecured = structurally subordinated to OpCo senior unsecured, EVEN IF both are labelled pari passu.

RATINGS IMPLICATIONS

Agencies notch HoldCo senior unsec 1–2 notches BELOW OpCo senior unsec.

When > 30% of group CFs are at subsidiary level and no downstream guarantee exists, the 2-notch adjustment is typical.

Investors ask: "What fraction of group debt sits at OpCo vs HoldCo? Is there a cross-guarantee? Are there dividend-restriction covenants at OpCo?"

Source: CFA Institute, Curriculum Vol 6, Learning Module 16, Practice Problem 8 (Mojofon holdco/opco), pp. 430–437.

SECTION 04

Securitization structures

Securitization moves an asset pool off the originator's balance sheet into a bankruptcy-remote SPV that issues ABS against those cash flows

The SPV/SPE is the critical legal device — "true sale" severs the asset from originator's bankruptcy estate. Without it, there is no securitization — just a secured bond.

1. **ORIGINATOR / SELLER**
   Creates or owns the assets (bank, auto-lender, credit-card issuer). Sells pool to SPV, receives cash.
2. **SPV / SPE / TRUST**
   Bankruptcy-remote legal entity that buys the pool, issues ABS, receives pool cash flows, pays bondholders.
3. **SERVICER**
   Administers loans: collects payments, chases delinquents, forecloses. Often the originator continues servicing.
4. **TRUSTEE**
   Independent fiduciary (usually a bank): holds pool cash, disburses to investors, reports performance monthly.
5. **RATING AGENCY**
   Rates each tranche on both collateral quality and seniority. Drives coupon pricing and investor eligibility.
6. **UNDERWRITER**
   Investment bank that structures the deal, markets to investors, provides liquidity in secondary.
7. **INVESTORS**
   Buy tranches matched to their mandate — pensions in senior AAA, hedge funds in mezz/equity.

Source: CFA Institute, Curriculum Vol 6, Learning Module 17, Exhibits 6–9, pp. 439–455.

Tranching stacks senior, mezzanine and equity bond classes on top of the same pool — each absorbs losses in reverse order of seniority

Arbitrage math: pool yield (MRR+250) minus weighted cost of tranches = residual spread going to equity. Equity earns this spread times leverage.

Tranche	Size ($)	Rating	Coupon	Attach	Detach	Expected loss	Role
Senior	70	Aaa / AAA	MRR + 40	30%	100%	~ 0 bp	Funding
Mezzanine	20	Baa / BBB	MRR + 250	10%	30%	~ 50 bp	Yield pick-up
Equity / residual	10	Unrated	Residual CF	0%	10%	First-loss	Issuer retains (skin in game)
Pool	**100**	**Mix**	**MRR + 250 (net of**	**–**	**–**	**–**	**–**

EQUITY ECONOMICS

Pool yield 250 bps over MRR. Weighted debt cost = (70×40 + 20×250) / 90 = 87 bps over MRR.

Residual to equity = (250 – 87) × 90/10 = 163 × 9 ≈ 14.7% excess return per unit of equity (before losses).

Source: CFA Institute, Curriculum Vol 6, Learning Module 17, Exhibit 3, pp. 443–445.

Loss-waterfall mechanics — a hypothetical 3-tranche deal shows how a 12% pool loss is absorbed entirely by equity and mezzanine, sparing senior

EUR 1,000 mm pool; tranches 700 / 200 / 100. Losses always absorbed bottom-up (equity first, then mezz, then senior).

Scenario	Pool loss (mm)	Equity (100)	Mezz (200)	Senior (700)	Sr recove
A — 5% pool loss	50	Loses 50 (50% hit)	Untouched (0)	Untouched (0)	1
B — 12% pool loss	120	Wiped (100)	Loses 20 (10% hit)	Untouched (0)	1
C — 25% pool loss	250	Wiped (100)	Wiped (200)	Loses 50 (7.1% hit)	92
D — 40% pool loss	400	Wiped (100)	Wiped (200)	Loses 100 (14.3% hit)	85

KEY INTUITION

Senior investors have a DEEP cushion. In scenarios A and B they take no loss at all — all risk has been transferred to the 300 mm of junior capital (30% subordination). Only when pool losses exceed 30% does senior start to suffer. That protection is why senior tranches can carry AAA ratings even when collateral is sub-investment-grade.

Source: CFA Institute, Curriculum Vol 6, Learning Modules 17–18; structure per Exhibit 3.

Credit enhancement reduces investor risk via internal (subordination, overcollateralization, reserve accounts) and external (letters of credit, guarantees, insurance wraps) mechanisms

Internal = built into the deal structure; external = third-party promises. Post-2008, external wrap providers are far less used — the monoline insurers collapsed.

INTERNAL

Subordination / tranching: junior bonds absorb losses before senior.

Overcollateralization (OC): pool face > bonds issued (e.g. 110 pool / 100 bonds). OC haircut is first-loss.

Reserve / cash-collateral account: cash set aside at closing to cover shortfalls.

Excess spread: pool coupon − bond coupons − fees — retained to absorb losses before OC triggers.

EXTERNAL

Letter of credit (LoC): bank commits to advance shortfalls up to a cap.

Third-party guarantee: parent / supranational / sovereign guarantees all bond payments.

Surety bond or monoline wrap: specialist insurer (Ambac, MBIA pre-2008) pledges to cover defaults.

Caveat: ABS investors are now exposed to the credit of the wrap provider — wrong-way risk in systemic stress (GFC 2008).

Source: CFA Institute, Curriculum Vol 6, Learning Module 18, pp. 463–480.

Non-mortgage ABS spans amortizing (auto loans) and non-amortizing (credit cards) collateral — the lockout / revolving period is the key distinction

During the lockout period, principal repaid on the pool is REINVESTED in new receivables. After lockout, amortization begins and principal flows to bondholders.

ABS type	Collateral	Amortizing?	Lockout period	Typical WAL	Key risk	AAA spread
Prime auto	New-car loans, 60–72m	Yes	0 (sequential)	1–2y	Unemployment	25–50
Subprime auto	Lower-credit auto loans	Yes	0	2–3y	Default, servicer	80–200
Credit-card	Revolving receivables	No (revolving)	2–5y	3–5y	Early amortization	35–60
Student loan (FFELP)	Govt-guaranteed	Yes	0	5–15y	Extension	40–80
Equipment lease	Aircraft, rail, trucks	Yes	0	3–7y	Obsolescence	60–150
						–250

EARLY AMORTIZATION TRIGGERS

Credit-card ABS include triggers that force EARLY AMORTIZATION if performance deteriorates (e.g. yield falls, charge-offs spike). Protects investors but shortens WAL unexpectedly — a hidden call option.

Source: CFA Institute, Curriculum Vol 6, Learning Module 18, pp. 482–500.

A CLO is the dominant CDO: an actively-managed pool of leveraged loans funded by a capital structure of debt tranches and an equity residual

Unlike a static ABS, the CLO manager actively trades the loan pool during a 4–5 year reinvestment period — performance can diverge from pool-level default experience.

Tranche	Typical size	Rating	Coupon	OC test
Class A (senior)	~ 65%	AAA	SOFR + 125–175 bps	Breach → cash diversion
Class B	~ 12%	AA	SOFR + 175–225 bps	Breach → cash diversion
Class C	~ 7%	A	SOFR + 225–275 bps	Breach → cash diversion
Class D	~ 6%	BBB	SOFR + 300–400 bps	Breach → cash diversion
Class E	~ 5%	BB	SOFR + 600–800 bps	Equity absorbed first
Equity (subordinated notes)	~ 5%	Unrated	Residual, 10–20% IRR	First-loss

CLO ARBITRAGE AND NON-LINEAR RISK

Arbitrage = weighted-average yield on leveraged-loan pool (~ SOFR + 350 bps) minus weighted-cost of debt tranches (~ SOFR + 200 bps). Non-linear risk: a small increase in default rates or drop in recovery can flip the OC test and cut off equity distributions entirely.

Source: CFA Institute, Curriculum Vol 6, Learning Module 18, pp. 498–505.

Covered bonds keep the cover pool on the ISSUER's balance sheet and give investors DUAL recourse — cover pool first, then general credit of the bank

European specialty — "Pfandbrief" in Germany since 1769. Adopted in Nordics, UK, France, Canada. Regulated by the EU Covered Bond Directive since 2019.

COVERED BOND (e.g. Pfandbrief)

Pool stays ON the issuer's balance sheet — "ring-fenced" but not sold.

Investors: dual recourse — first on ring-fenced pool; if insufficient, claim on bank's general assets.

Overcollateralization: collateral value always > face of bonds issued (legal minimum, e.g. 2–5% excess).

Only financial institutions issue; only eligible collateral (prime residential mortgages < 80% LTV typically).

TRUE-SALE ABS (e.g. RMBS)

Pool is SOLD to a bankruptcy-remote SPV — legal true-sale transfer.

Investors: recourse ONLY to pool and its enhancements. No claim on the originator.

Tranching creates multiple bond classes at different ratings. Credit enhancement is primarily INTERNAL.

Any corporate (bank or non-bank) can sponsor an ABS. Wide collateral universe.

Source: CFA Institute, Curriculum Vol 6, Learning Module 18, pp. 464–472.

SECTION 05

MBS specifics

Mortgage-backed securities divide along two axes: residential vs commercial, and agency (GSE-backed) vs non-agency

Agency MBS = Ginnie Mae (full faith & credit of US Treasury), Fannie Mae, Freddie Mac (implicit/explicit US gov support since 2008 conservatorship).

	Agency (US GSE)	Non-agency (private label)
RMBS	Conforming loans only (< $726k 2023, 80% LTV, FICO ≥ 620). Credit risk absorbed by GSE.	Jumbo, subprime, Alt-A, non-QM loans. Credit risk retained by structure (tranching + enhancement).
CMBS	Limited — Ginnie Mae multifamily project loans; Fannie/Freddie multifamily.	Commercial/multifamily pools: offices, retail, hotels, industrial. Typically conduit or single-asset / single-borrower.

AGENCY RMBS

Credit risk eliminated (GSE guarantee). Investor bears ONLY prepayment risk + interest-rate risk. Most liquid MBS market in the world; TBA trading convention.

NON-AGENCY RMBS

Investor bears credit risk + prepayment + rate. Ratings driven by pool LTV, FICO, debt-to-income, doc quality. Post-2008: far smaller market, tighter underwriting.

CMBS

Investor bears credit risk but benefits from call protection (defeasance, yield maintenance, prepayment lockout). Balloon risk replaces contraction risk.

Source: CFA Institute, Curriculum Vol 6, Learning Module 19, Exhibit 1–2, pp. 515–525.

Pass-through pool characteristics are summarised by its weighted-average coupon (WAC), weighted-average maturity (WAM), and pass-through rate

Pass-through rate = WAC − servicing fee − guarantee fee. Typical: 50 bps total spread between WAC and coupon received by MBS holder.

$$WAC = \Sigma\, w_i \times c_i \qquad WAM = \Sigma\, w_i \times m_i \qquad w_i = Balance_i \,/\, Total$$

where: c_i = mortgage i coupon; m_i = remaining months to maturity; w_i = balance weight. Pool is re-weighted monthly as loans amortize and prepay.

Mortgage	Balance ($)	Weight	Coupon	Rem. term (months)	w × c	w × m
A	125,000	12.5%	2.50%	240	0.313%	30.0
B	340,000	34.0%	3.30%	288	1.122%	97.9
C	74,000	7.4%	2.80%	180	0.207%	13.3
D	132,000	13.2%	4.00%	240	0.528%	31.7
E	329,000	32.9%	3.70%	252	1.217%	82.9

PASS-THROUGH RATE

WAC = 3.29%, WAM = 255.8 months (21.3y). If servicing fee = 25 bps and guarantee fee = 6 bps, pass-through rate = 3.29% − 0.31% = 2.98%.

Source: CFA Institute, Curriculum Vol 6, Learning Module 19, Exhibit 4, pp. 525–535.

Prepayment risk is the defining risk of residential MBS — it has two faces: contraction risk when rates fall (refi wave) and extension risk when rates rise

Mortgage investors are implicitly SHORT a prepayment option. When they benefit (rates up, bond price shouldn't fall), borrowers slow prepay — and vice versa.

CONTRACTION RISK (rates ↓)

When mortgage rates fall, homeowners refinance into lower-rate loans.

→ Prepayments accelerate; pool WAL shortens.

→ Investor receives principal early — exactly when reinvestment rates are LOWER.

→ Price appreciation capped (negative convexity): unlike a normal bond that rallies fully, MBS rally is truncated by accelerating prepay.

EXTENSION RISK (rates ↑)

When mortgage rates rise, homeowners stay in their below-market loans.

→ Prepayments slow; pool WAL lengthens.

→ Investor is locked into a LOW-coupon bond while market yields are higher.

→ Price falls MORE than a comparable callable bond — duration extends just when you want it shorter.

Source: CFA Institute, Curriculum Vol 6, Learning Module 19, pp. 535–545.

Single monthly mortality (SMM) is the fraction of pool principal prepaid in a month; the PSA benchmark describes the ramp of conditional prepayment rate (CPR) over a pool's life

PSA benchmark: CPR of 0.2% in month 1, ramps +0.2% per month until month 30, then flat 6% CPR. "100 PSA" = 100% of benchmark speed. "150 PSA" = 1.5× benchmark.

$$SMM = \text{Prepay} / (\text{Beginning balance} - \text{Scheduled principal})$$

$$CPR = 1 - (1 - SMM)^{12} \qquad \text{PSA speed} \times 100 \text{ PSA benchmark} = \text{effective CPR}$$

where: SMM = monthly; CPR = annualised conditional prepayment rate; PSA benchmark slopes from 0.2% → 6% CPR over months 1–30 then flat.

Month	100 PSA CPR	150 PSA CPR	SMM at 150 PSA	250 PSA CPR	SMM at 250 PSA
1	0.20%	0.30%	0.025%	0.50%	0.042%
12	2.40%	3.60%	0.305%	6.00%	0.514%
30	6.00%	9.00%	0.783%	15.00%	1.340%
60	6.00%	9.00%	0.783%	15.00%	1.340%
120	6.00%	9.00%	0.783%	15.00%	1.340%

Source: CFA Institute, Curriculum Vol 6, Learning Module 19, pp. 540–552.

150 PSA on a $100mm 30-year pool at month 36 means CPR 9.00%, SMM 0.783%, and $782,600 of voluntary prepayment on top of scheduled principal

Month 36 is well past the PSA ramp (which ends at month 30). So CPR is flat at 9%, SMM flat at 0.783%.

CASH-FLOW BUILD — $100mm 30y POOL @ 4% WAC, MONTH 36, 150 PSA

Beginning balance = $100,000,000. 150 PSA means CPR = 9%. SMM = 1 − (1 − 0.09)^(1/12) = 0.00783 = 0.783%.

1. (1) Interest CF = Balance × WAC / 12 = 100,000,000 × 4% / 12 = $333,333
2. (2) Scheduled principal ≈ balance × factor ≈ 100,000,000 × 0.00167 ≈ $167,000 (from amortization table)
3. (3) Prepay base = Balance − Scheduled principal = 100,000,000 − 167,000 = $99,833,000
4. (4) Voluntary prepayment = SMM × Prepay base = 0.00783 × 99,833,000 = $781,692
5. (5) Total principal CF = Scheduled + Prepay = 167,000 + 781,692 = $948,692
6. (6) Total cash flow = Interest + Principal = 333,333 + 948,692 = $1,282,025

Pool balance falls to $99,051,308 at start of month 37. Pool WAL is ~7.5y at 150 PSA vs 10y at 100 PSA.

Source: CFA Institute, Curriculum Vol 6, Learning Module 19, PSA worked examples, pp. 545–555.

A sequential-pay CMO redirects all pool principal to Tranche A first, then B, then Z — shortening A's WAL and lengthening Z's

Pass-through alone has single WAL for all investors. Tranching creates a menu — short-WAL tranche A for money-market buyers, long-WAL Z for liability-matched insurers.

Tranche	Size	Priority	WAL @ 165 PSA	WAL @ 250 PSA	Target buyer
Tranche A (short)	$200mm	All principal until paid off	3.3y	2.2y	Money-market
Tranche B (intermediate)	$140mm	Principal after A fully repaid	7.5y	5.3y	Insurer
Tranche Z (accrual)	$60mm	Interest accrues; principal last	22.0y	16.5y	Pension
Pool total	**$400mm**	**–**	**10.0y**	**6.8y**	**–**

WHY SEQUENTIAL-PAY WORKS

Tranche A absorbs ALL prepayment risk first — its WAL variability is still high but the absolute maturity is short, so price volatility is small. Tranche Z absorbs very little contraction risk (by the time A and B are paid off, most prepayment waves are over) so its CF is more predictable. Investors pick the tranche matching their duration needs.

Source: CFA Institute, Curriculum Vol 6, Learning Module 19, pp. 555–568.

A planned-amortization-class (PAC) structure guarantees a schedule of principal payments over a collar of prepayment speeds — support tranches absorb the volatility

PAC collar e.g. 100–250 PSA. Inside the collar, PAC CF is certain. Outside: if too fast, support absorbs excess; if too slow, support delays principal to PAC.

PAC TRANCHE

Receives a scheduled principal payment each month.

Inside collar (e.g. 100–250 PSA): schedule honoured EXACTLY.

Above upper bound: support tranche buffers the excess prepayment.

Below lower bound: support's principal is redirected TO the PAC — only after prolonged slow prepay does PAC schedule break.

Result: narrow band of WAL uncertainty; tight spread; popular with insurers & pensions.

SUPPORT TRANCHE

Absorbs ALL prepayment variability around the PAC schedule.

Above upper bound: extra principal flows to support (CONTRACTION — WAL shortens).

Below lower bound: support gets no principal until PAC is caught up (EXTENSION — WAL lengthens).

Result: support WAL is extremely variable; spread 50–150 bps wider than comparable non-structured MBS.

Only buyers: specialists with appetite for convexity risk.

Source: CFA Institute, Curriculum Vol 6, Learning Module 19, pp. 568–575.

CMBS differs from RMBS: commercial mortgages are non-recourse, usually have call protection, and frequently include a balloon — so balloon risk replaces contraction risk

RMBS investor worries about prepayment. CMBS investor worries about loan-level credit, concentration, and inability to refinance the balloon at maturity.

Dimension	RMBS	CMBS
Recourse	Full (most jurisdictions)	Typically non-recourse — lender relies on property cash flow
Pool size	Thousands of small loans	20–200 large loans; single-asset CMBS exists
Concentration risk	Low (diversified)	High — one large loan can be 5–15% of deal
Call protection	None — borrower may prepay anytime	Lockout + defeasance / yield-maintenance penalty
Primary risk	Prepayment (contraction + extension)	Balloon refinance risk + credit risk

BALLOON RISK

Most commercial mortgages are 10-year interest-only or partially amortizing with a balloon maturity. At balloon, borrower must refinance or sell — if refinance markets are closed (2008–09, 2023 office) borrower defaults despite positive cash flow. CMBS investor is exposed to this binary refinancing event.

Source: CFA Institute, Curriculum Vol 6, Learning Module 19, pp. 575–585.

KEY TAKEAWAYS

Deck 16 — Fixed Income IV — Credit Analysis & Securitization

- Credit risk decomposes into POD × LGD = expected loss. A fair credit spread approximates EL; LGD = EE × (1 − RR).
- The investment-grade / high-yield line sits at Baa3 (Moody's) and BBB− (S&P, Fitch); crossing it triggers mandate-driven forced selling.
- Sovereign creditworthiness = ability to pay (taxation, fiscal strength, FX reserves) AND willingness to pay (sovereign immunity limits enforcement).
- Corporate credit rests on 4 C's (Capacity, Capital, Collateral, Covenants, Character) plus 3 top-down (Country, Conditions, Currency), driving profitability, coverage, leverage ratios.
- Priority of claims cascades first lien → senior secured → senior unsecured → subordinated → equity; recovery ranges from ~65% (first lien) to 0% (equity).
- Securitization: true-sale transfer to a bankruptcy-remote SPV, tranching redistributes risk; senior AAA typically has 30%+ subordination cushion.
- Credit enhancement is internal (subordination, OC, reserve accounts, excess spread) or external (LoCs, guarantees, insurance wraps).
- Residential MBS investors are short a prepayment option; 150 PSA on a 30y pool post-month-30 means 9% CPR / 0.78% SMM. CMBS trades balloon risk for prepayment risk via call protection.